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System	A system is one which operates on an input to produce an output. – electrical, mechanical, hydraulic, pneumatic, biomedical etc variables like temperature, pressure, liquid level etc Examples of systems are amplifier in electronic engineering, process in chemical engineering, antenna tracking system in electrical engineering etc Each system produces some output for a given input.
Control system	Control system is an arrangement of physical components that monitor and command a system so that a desired objective is obtained.

TYPES OF CONTROL SYSTEMS

OPEN LOOP CONTROL SYSTEMS (NON-FEEDBACK SYSTEMS)

An open loop control system utilizes an actuating device to control the process without feedback.

An open-loop control system is one in which the output is dependent on input, but the input is totally independent of the output or changes in output of the system.

In an open loop system, the output can be adjusted / varied by varying the input. There is no feedback effect on the input. The output of the open loop system can be determined only by the input. If the output is affected due to some external disturbance, the open loop system does not correct it. In open loop systems, the changes in output are corrected by changing the input manually.

The block diagram of an open-loop system is shown below.



Example of open loop control system	Electric clothes dryer: Depending upon the amount of clothes to be dried, an operator would set a timer (controller) to say 30 minutes. After the 30 minutes the drier will automatically stop and turn-off. The clothes are then still wet or damp. Here, the clothes dryer is an open-loop control system. It does not monitor or measure the condition of the output signal, but it only depends on the input.
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Advantages of open loop system	variables to provide feedback.
	Economical
	Easy from maintenance point of view
	No stability problem. Generally, they are stable.
	These systems are convenient to use when output is difficult to measure.

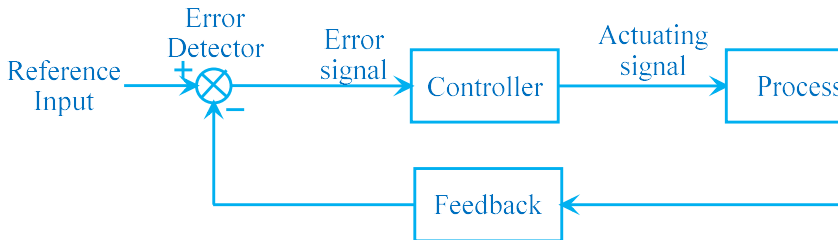
Disadvantages of open loop system	Not accurate and reliable because their accuracy is dependent on the quality of the components.
	The changes in the output due to external disturbances are not corrected automatically.
	Recalibration of the controller is required from time to time to maintain the quality and accuracy.

CLOSED LOOP CONTROL SYSTEMS (FEEDBACK SYSTEMS)

Control systems in which the output has an effect upon the input quantity in order to achieve the desired output value are called closed-loop control systems.

An open-loop control system can be modified into a closed-loop control system by introducing feedback.

The general block diagram of an automatic control system is shown in the figure below.



The reference signal (or input signal) corresponds to the desired output. The output is measured and converted to a feedback signal of the same type as that of the reference signal. This feedback signal is proportional to the output signal and it is fed to the error detector. The error signal is the difference between the reference signal and the feedback signal. The error detector is the difference between the reference signal and the feedback signal.

control system	
	Air Conditioner: The air conditioner measures the actual room temperature, compares it with the desired input temperature, and controls the room temperature by turning the compressor ON or OFF according to the requirement. Traffic-light system: Some traffic-light systems are closed-loop systems. In the open-loop type, there is a timing mechanism that turns the lights at regular intervals, irrespective of the volume of traffic. In the closed-loop type, however, the amount of traffic passing through the junction is monitored (electronically or otherwise), and the duration of the lights is adjusted for each part of the junction according to the conditions.

Advantages of closed loop system	More accurate than open loop systems because of the presence of feedback.
	Closed loop systems reduce the effect of noise and disturbance on the system performance.
	The sensitivity of the closed-loop systems for parameter variations is reduced by increasing the feedback loop gain.
	The range of frequencies over which the system responds is increased, resulting in the increased bandwidth.
	If an open-loop system is unstable, it is possible to make it stable by providing feedback.
	There is reduced effect of non-linearities in these systems.

Disadvantages of closed loop system	Closed-loop systems are complicated in design.
	Closed-loop systems are costlier.
	Feedback reduces the overall gain of the system.
	Stability is a major problem in closed loop system.

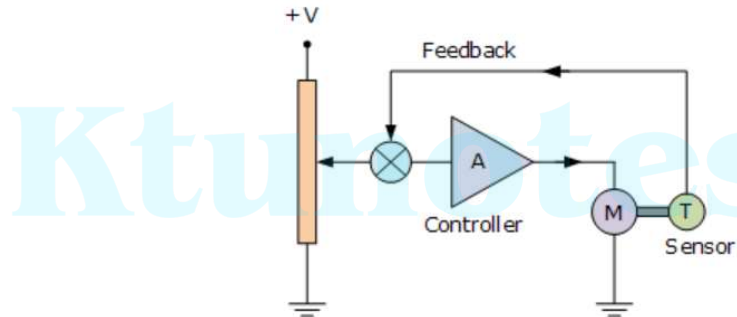
It has less maintenance

It has more maintenance

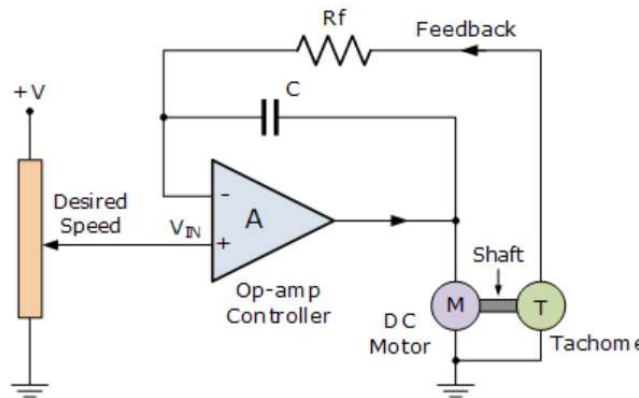
Example: Washing machine, room heater

Example: refrigerator, serv

CLOSED LOOP SPEED CONTROL OF DC MOTOR



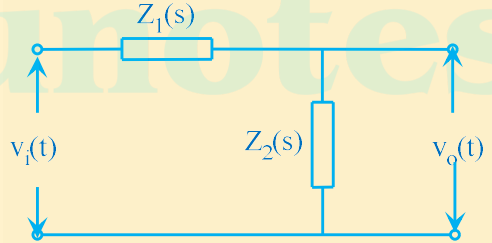
The position of the potentiometer slider represents the reference input. A tachogenerator connected to the shaft produces a feedback signal proportional to the motor speed. The difference between the input voltage setting and the feedback voltage level gives the error signal. An external disturbance like change in load will produce an error signal which the system automatically responds to by adjusting the motor speed. The controller works to minimize the error, leading to zero error which indicates actual speed equal to set point.



$$G(s) = \frac{\text{Laplace transform of output}}{\text{Laplace transform of input}} \bigg|_{\substack{\text{all initial conditions} \\ \text{are zero}}}$$

$$G(s) = \frac{C(s)}{R(s)} \bigg|_{\substack{\text{all initial conditions} \\ \text{are zero}}}$$

Determine the transfer function of the electrical network shown in figure.

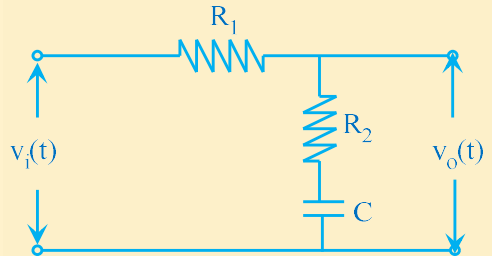


$$I(s) = \frac{V_i(s)}{Z_1(s) + Z_2(s)}$$

$$V_o(s) = Z_2(s)I(s) = \frac{Z_2(s)V_i(s)}{Z_1(s) + Z_2(s)}$$

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

Determine the transfer function of the phase-lag network shown in figure.

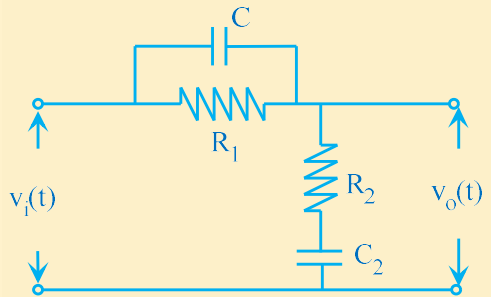


$$Z_1(s) = R_1 // \frac{1}{sC_1} = \frac{R_1 \times \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{R_1}{sR_1C_1 + 1}$$

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)} = \frac{R_2}{\frac{R_1}{sR_1C_1 + 1} + R_2} = \frac{R_2(sR_1C_1 + 1)}{R_1 + R_2(sR_1C_1 + 1)}$$

$$G(s) = \frac{R_2(sR_1C_1 + 1)}{R_1 + R_2(sR_1C_1 + 1)} = \frac{R_2(sR_1C_1 + 1)}{R_1 + R_2 + sR_1R_2C_1}$$

Determine the transfer function of the lag-lead compensator network shown in



$$Z_1(s) = R_1 // \frac{1}{sC_1} = \frac{R_1 \times \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{R_1}{sR_1C_1 + 1}$$

$$Z_2(s) = R_2 + \frac{1}{sC_2} = \frac{sR_2C_2 + 1}{sC_2}$$

FEEDBACK CONTROL

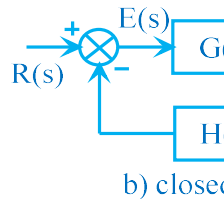
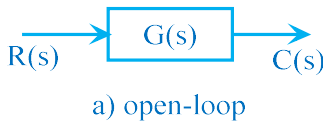
The feedback control is an operation in which the output is sampled and a feedback to the input. The feedback output may be in phase or out-of-phase with the input. When the feedback output is in phase with the input it is termed positive feedback. When the feedback output is out-of-phase with respect to the input it is termed negative feedback.

All control systems are usually negative feedback systems. In a negative feedback system, the difference between the reference input and the output produces an error which is reduced to bring the output of the system to a desired level.

The positive feedback output gets added to the reference input and increases the output to instability. But, sometimes, positive feedback is used in minor loops to amplify certain internal signals or parameters. (Positive feedback increases the output to instability in a system which may lead to self-oscillation and as such, positive feedback is used in oscillatory circuits such as Oscillators and Timing circuits)

EFFECTS OF FEEDBACK

The error between the system input and output can be reduced by using a feedback



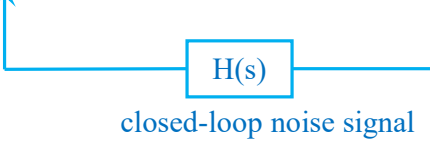
Here, $G(s)$ = open-loop gain

The overall transfer function of the closed-loop system is derived as follows: -

$$R(s) - C(s)H(s) = E(s)$$

$$\text{But, } C(s) = E(s)G(s) \text{ ----- (1)}$$

Effect of feedback on overall gain	If the value of $(1+GH)$ is less than 1, then the overall gain of the system is less than the open loop gain. In this case, GH is negative because the gain of the feedback path is negative. If the value of $(1+GH)$ is greater than 1, then the overall gain of the system is greater than the open loop gain. In this case, GH is positive because the gain of the feedback path is positive. In general, G and H are functions of frequency. So, the overall gain of the system in one frequency range and another frequency range.
Effect of feedback on stability	A system is said to be stable, if its output is under control and does not diverge to infinity. If the system is not stable, it is said to be unstable. $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$ If the denominator value is zero (i.e., $GH = -1$), then the overall gain of the system will be infinite. So, the control system becomes unstable. Therefore, we have to properly choose the feedback path to make the control system stable.
Effect of feedback on sensitivity	In general, a good control system should be very insensitive to variations but sensitive to input commands. Sensitivity of the overall gain of closed loop control system is defined as the ratio of the change in overall gain to the change in open loop gain of $(1+GH)$. So, Sensitivity may increase or decrease depending on the value of $(1+GH)$. If the value of $(1+GH)$ is less than 1, then sensitivity is high. If GH is negative because the gain of feedback path is negative. If the value of $(1+GH)$ is greater than 1, then sensitivity is low. If GH is positive because the gain of feedback path is positive. In general, 'G' and 'H' are functions of frequency. So, the sensitivity of the system gain in one frequency range and another frequency range. Therefore, we have to choose the feedback path in a way that the system is insensitive or less sensitive to variations.
Effect of feedback on external disturbance or noise	To know the effect of feedback on noise, let us compare the system relations with and without feedback due to noise signal. Consider an open loop control system with noise signal.

	 closed-loop noise signal The closed loop transfer function due to noise is given by $\frac{C(s)}{N(s)} = \frac{G_b(s)}{1 + G_a(s)G_b(s)H(s)}$.It is obtained by making the denominator equal to zero. In the closed loop control system, the gain due to noise is reduced by a factor of $(1 + G_aG_bH)$ provided that the term $(1 + G_aG_bH)$ is not equal to zero.
Effect of feedback on bandwidth, impedance, transient response and frequency response	Feedback can affect performance characteristics such as bandwidth, impedance, transient response and frequency response.

TYPE AND ORDER OF A SYSTEM

The transfer function can be written in any of the following forms:

1	$\frac{C(s)}{R(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$	The numerator is a polynomial of degree m and the denominator is a polynomial of degree n. The degree n of the denominator is called order of the system.
2	$\frac{C(s)}{R(s)} = \frac{K(1 + sT_{z1})(1 + sT_{z2})\dots(1 + sT_{zm})}{(1 + sT_{p1})(1 + sT_{p2})\dots(1 + sT_{pn})}$	Time-constant
3	$\frac{C(s)}{R(s)} = \frac{K'(s - z_1)(s - z_2)\dots(s - z_m)}{(s - p_1)(s - p_2)\dots(s - p_n)}$	Pole-zero $z_1, z_2, \dots, z_m$ = Roots of numerator polynomial = zeros

$G(s)H(s)$ = open-loop transfer function

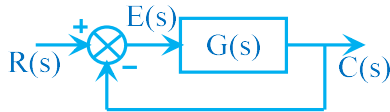
loop system

General representation of open-loop transfer function $G(s)H(s)$ of a feedback system

$$G(s)H(s) = \frac{K(1+sT_{z1})(1+sT_{z2})\dots\dots(1+sT_{zm})}{s^N(1+sT_{p1})(1+sT_{p2})\dots\dots(1+sT_{pn})}$$

Here, N is the number of poles at the origin and it represents the **type** number of the system

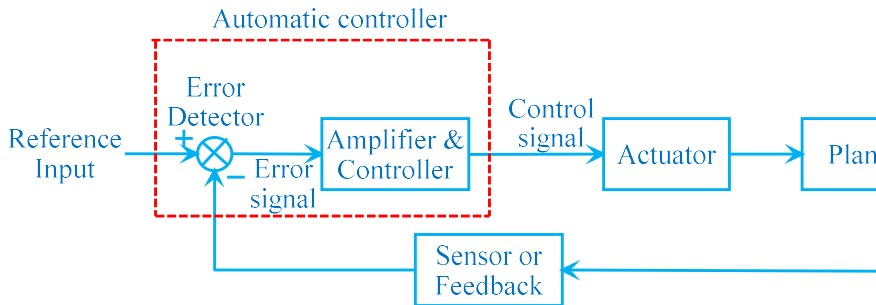
The **type** of the system is defined as the number of poles at the origin of the open-loop transfer function $G(s)H(s)$.



Note: For a unity feedback system, $H(s) = 1$, hence open-loop transfer function is $G(s)$.

CONTROL SYSTEM COMPONENTS

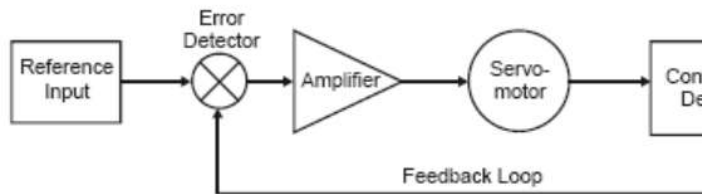
The basic components of an automatic control system are error detector, amplifier, actuator, plant and sensor or feedback system. The block diagram of an automatic control system is shown below.



		error signal.
	Controller	The controller itself amplifies the error signal, differentiates it to produce a control signal. The controllers are P, PI, PD and PID controllers.
	Actuator	The actuator is a power amplifying device that produces motion in the plant according to the control signal. The actuator is a motor. Examples of electric motors employed are servomotor, AC servomotor and stepper motor.
	Sensor or feedback system	The feedback system samples the output to produce a feedback signal which is proportional to the current output. The sensor converts the output variable into another suitable form such as displacement, pressure or voltage so that it can be fed back to the reference input. Tacho-generators, transducers are used in feedback systems.

SERVOMOTOR

A servo motor is a rotary actuator or linear actuator that allows for precise control of its position, velocity, and acceleration. Block diagram of a typical servo system is shown below.



A reference input (typically called a velocity input) is sent to the servo amplifier. The amplifier controls the speed of the servomotor. Directly mounted to the servomotor is a feedback device that converts mechanical motion into electrical signals and is used as a feedback signal. This signal is sent to the error detector and is compared with the reference input. If there is a difference, the error signal is sent directly to the amplifier, which makes the necessary corrections.

CONSTRUCTION OF AC SERVOMOTOR: Similar to a normal induction motor, it comprises of a stator and rotor.

STATOR: The stator of this motor which has a laminated structure is wound with two windings placed 90° electrical apart in space. One of the windings called the main winding is connected to a constant-voltage supply source. The other winding called the control winding is connected to a variable control voltage which is 90° electrical out of phase with respect to the main winding. The control voltage is supplied from a servo amplifier.

ROTOR: The rotor is usually of the squirrel cage type with small diameter and low mechanical inertia as low as possible. It has high resistance in order to have a linear characteristic as linear as possible. The inertia of the rotor is further reduced for very low power applications. This type of rotor is a special form of squirrel cage rotor. The conductors are in the form of a drag-cup made of a non-magnetic material such as copper or an alloy. The sotted rotor laminations are replaced by a set of stationary ring laminations to provide a low-reluctance path for the magnetic flux.

To minimize rotor kinetic energy, servo motors are wound with as many poles as possible to operate at low speeds. This also provides corresponding increase in torque. AC servomotors have air gaps to reduce magnetizing current and consequent losses.

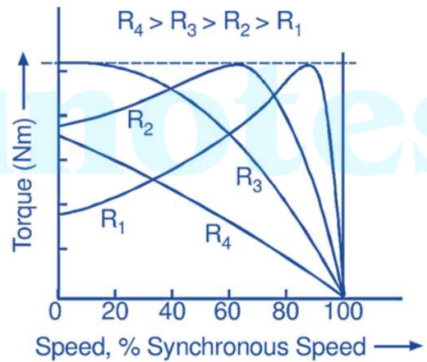
PRINCIPLE OF OPERATION: One phase, designated as reference phase, is connected to a constant voltage source, the frequency of which is usually 50/60 or 400Hz. The second phase, designated as the control phase, is excited by a voltage of variable magnitude and frequency. The control voltage is obtained from the servo amplifier. It is to be noted here that to have a rotating magnetic field, the voltage applied to the control winding must be 90° out of phase w.r.t. the reference voltage. The required phase difference is provided by a capacitor. Rotating magnetic field induces currents in rotor conductors. The rotating magnetic field interacts with the rotor currents to produce a torque on the rotor in the direction of field rotation. This causes the rotor to rotate.

TRANSFER FUNCTION OF AC SERVOMOTOR:

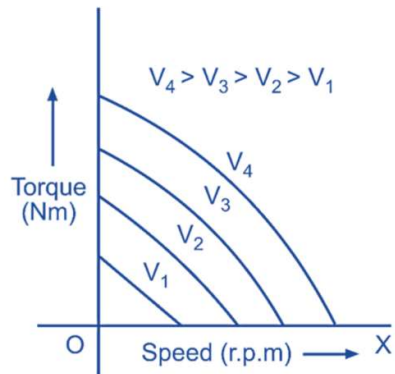
$$T = K_1 e_c - K_2 \frac{d\theta}{dt}$$

$$T(s) = K_1 E_c(s) - K_2 s \theta(s)$$

TORQUE–SPEED CHARACTERISTICS: The general shape of the torque-speed characteristics of a two-phase induction motor depends largely on the value of rotor resistance. Rotor resistance linearizes the torque-speed characteristics. Because of this, the rotor is built with high resistance so that its torque-speed characteristics is nearly linear.



Torque-speed characteristics of a servomotor for various control voltages (V_1 , V_2 , V_3 , V_4) are shown in figure below.



It is observed that the torque (particularly starting torque) varies almost linearly with control voltage. The torque for any particular value of control voltage is

DC SERVOMOTOR

Except some minor differences in constructional features, a DC servo motor is a DC motor. To meet the requirement for low inertia, DC servomotors are designed with a small diameter to length ratio for their armatures. Further, to attain linear torque-speed characteristics, they are generally separately excited.

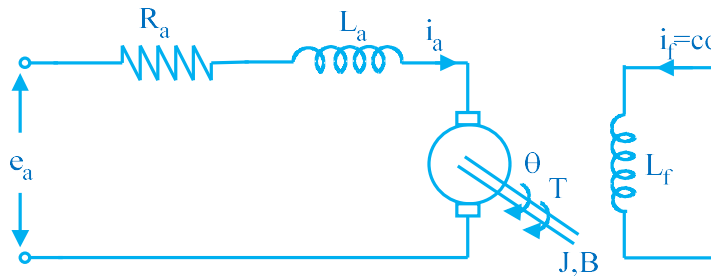
DC servomotors can be controlled either by field current or armature current. They are classified as i) armature controlled DC servomotor or ii) field controlled DC servomotor.

ARMATURE CONTROLLED DC SERVOMOTOR

In this case, the output (ie control signal) of the servo-amplifier is applied to the armature circuit. The field current is kept constant. In comparison with field control, armature control gives better linear performance with rapid response. For large size motors, use of armature control is more economical.

TRANSFER FUNCTION OF ARMATURE CONTROLLED DC MOTOR

Representation of an armature-controlled DC motor is shown in figure. An external source supplies a constant current i_f to the field winding. Control signal is applied to the armature circuit.



Let

R_a = armature winding resistance

L_a = armature winding inductance

e_a = applied armature voltage

e_b = back emf

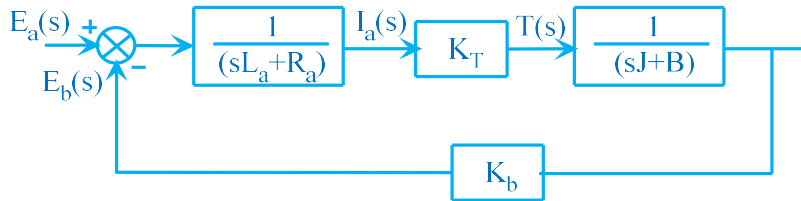
$$\text{But, } I_a(s) = \frac{T(s)}{K_T} = \frac{\theta(s)(Js^2 + Bs)}{K_T}$$

$$E_a(s) = \frac{\theta(s)(Js^2 + Bs)}{K_T} (R_a + sL_a) + K_b s \theta(s)$$

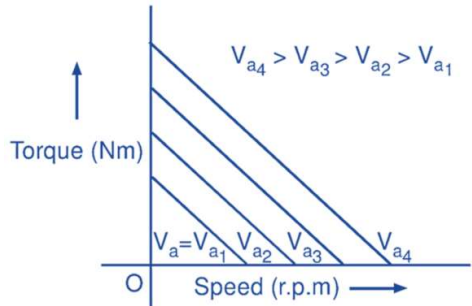
$$\frac{\theta(s)}{E_a(s)} = \frac{K_T}{s(Js + B)(R_a + sL_a) + K_T K_b s} = \frac{\frac{K_T}{(Js + B)(R_a + sL_a)}}{1 + \frac{K_T K_b}{(Js + B)(R_a + sL_a)}} \times \frac{1}{s}$$

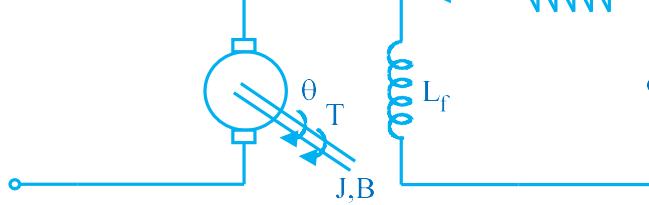
$$\text{This is of the form, } \frac{\theta(s)}{E_a(s)} = \frac{G}{1 + GH} \times \frac{1}{s}$$

Hence, block diagram of armature-controlled DC motor is given by,



TORQUE-SPEED CHARACTERISTICS





$$e_f = i_f R_f + L_f \frac{di_f}{dt}$$

$$T = K_T' i_f$$

$$T = J \frac{d^2 \theta}{dt^2} + B \frac{d\theta}{dt}$$

$$E_f(s) = I_f(s) (R_f + sL_f)$$

$$E_f(s) = I_f(s) (R_f + sL_f)$$

$$T(s) = K_T' I_f(s)$$

$$T(s) = \theta(s) (Js^2 + Bs)$$

$$\text{But, } I_f(s) = \frac{T(s)}{K_T'} = \frac{\theta(s) (Js^2 + Bs)}{K_T'}$$

$$E_f(s) = \frac{\theta(s) (Js^2 + Bs)}{K_T'} (R_f + sL_f)$$

$$\frac{\theta(s)}{E_f(s)} = \frac{K_T'}{s (Js + B) (R_f + sL_f)}$$

Block diagram of a field-controlled DC motor is shown below



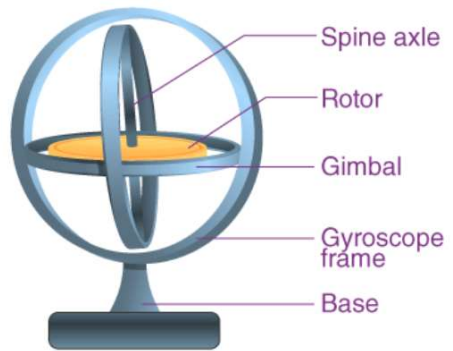
Even though the power required for field control is only a fraction of the power control, it is less common. This is because the time constant of the field circuit (L_f/R_a) with the armature circuit time constant (L_a/R_a). Consequently, field control response as armature control. Also, the torque-speed characteristics under field

It includes permanent magnet in its construction	The synchronous type A permanent magnet while i require it.
These motors are used for high power applications	These motors are use applications

GYROSCOPE

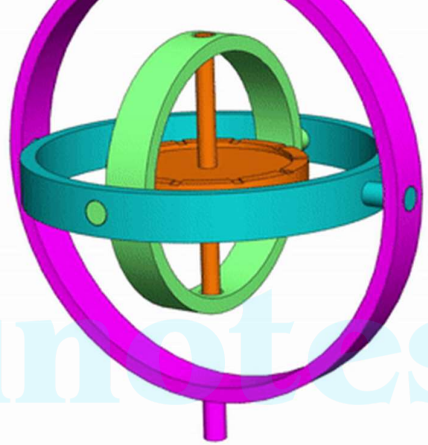
Gyroscopes are devices mounted on a frame and able to sense an angular velocity

They are classified into mechanical gyroscopes; optical gyroscopes, micro-electromechanical (MEMS) gyroscopes.



A mechanical gyroscope essentially consists of a spinning mass that rotates around a fixed axis. When the mass is rotating on its axis, it tends to remain parallel to itself and to resist any change in its orientation. If a gyroscope is installed on gimbals that allow the mass to rotate in the three directions of space, its spinning axis will remain oriented in the same direction even if the support changes direction.

The primary application of gyroscopic effects consists in the measurement of the angular velocity of a moving vehicle. The spinning mass is mounted upon a gimbaled frame, allowing it to rotate about three perpendicular axes. The gimbaled frame of the gyroscope is attached to the vehicle, which can rotate, while the rotation axis of the spinning mass keeps its angular position fixed relative to the inertial space.



Applications of Gyroscope

1. Gyroscopes find applications in the compasses of boats, spacecraft orientation and the pitch of the aeroplane are determined against a gyroscope.
2. In spacecraft, the navigation of the desired target is done with the help of the spinning centre of the gyroscope is used as the orientation point.
3. The stabilization of the large boats and satellites is done with the help of a gyroscope.
4. Gyroscopes are used in gyrotheodolites for the maintenance of the direction of a line.
5. Gyroscopes along with accelerometers are used in the design of a system for excellent motion sensing.

DC TACHOGENERATOR

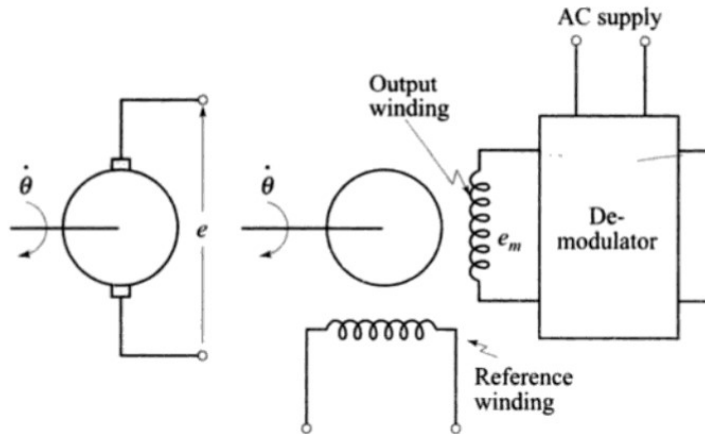
A DC tachogenerator consists of a small armature which is coupled to the motor whose speed is to be measured. This armature revolves in the field of a permanent magnet. The induced voltage is proportional to the product of flux and speed. Since the flux of the permanent magnet is constant, the voltage generated is proportional to the speed. The polarity of the output voltage is determined by the direction of rotation. Commutator and brushes are used to convert the alternating current into a direct current.

$$\frac{E(s)}{\theta(s)} = sK_t$$

APPLICATIONS: Tachometer generators are often utilized to gauge engine speed with the matching rate of powered equipment such as conveyors, mixers, fans and other devices. They are consequently used as feedback elements in speed control circuits.

AC TACHOGENERATOR

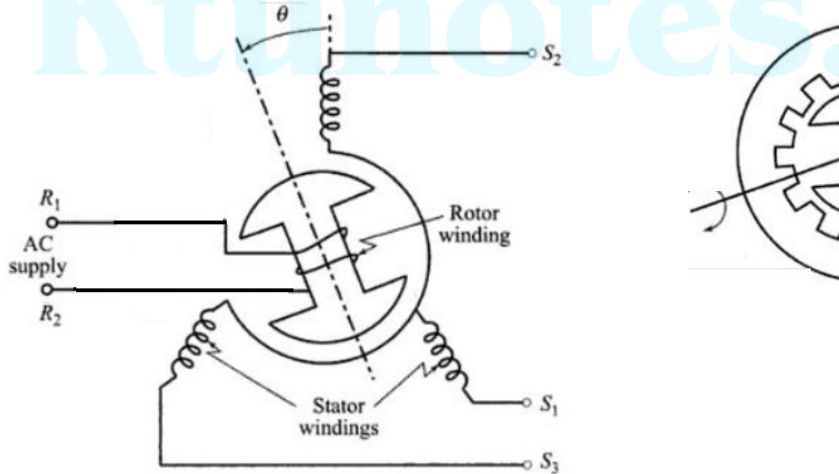
AC tachogenerator resembles a two-phase induction motor. It comprises two stator windings (one as reference winding and output winding) arranged in space quadrature and a rotating rotor. The reference winding is excited by a sinusoidal voltage with frequency ω_c . When the rotor is stationary, no emf is induced in the output winding. When the rotor rotates, an emf will be induced in the output winding. Magnitude of the output voltage will be proportional to the rotational speed. The direction of shaft rotation causes a 180° phase shift in the output voltage.



If the reference winding is excited by the voltage $e_r(t) = E_r \sin \omega_c t$, an emf of magnitude e_m will be induced in the output winding if the rotor is rotating. The output voltage of the tachogenerator is thus, in a modulated form. The information about shaft speed is stored not in the magnitude of the output voltage $e(t)$ is separated by using a demodulator (signal conditioner).

1. Synchro Transmitter
2. Synchro receiver

SYNCHRO TRANSMITTER –It has a stator and rotor. The stator of the synchro is designed to reduce the iron losses. Stator winding is made up of conductors insulated from each other and placed in the slots of the stator core. There are three windings on the stator side, they are physically displaced by 120° from each other around the periphery. The rotor is made of sheet steel stampings with good magnetic property. Rotor core may be of different shapes (pole type), umbrella shaped or drum-shaped cylindrical rotor.



The single-phase AC voltage is applied to the rotor with the help of slip rings and brushes. The current produces an alternating flux in the air-gap and induces voltage in the stator windings by mutual induction. The induced voltage in the stator coil depends on the angular position of the rotor with respect to the rotor axis.

Let $e(t) = E \sin \omega t$ is the AC voltage is applied to the rotor of the transmitter. When the rotor is in the position corresponding to $\theta = 0$, the voltage induced across the stator winding is maximum and is written as

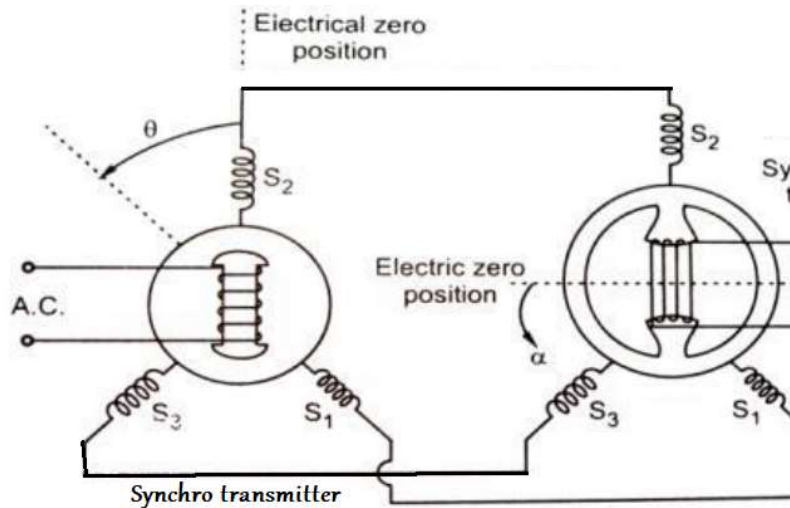
$$e_{S_2n}(t) = KE \sin \omega t \text{ where } K \text{ is a proportionality constant.}$$

$$e_{S_3S_1} = e_{S_3n} - e_{S_1n} = \sqrt{3}KE \sin \theta \sin \omega t$$

SYNCHRO CONTROL TRANSFORMER: The constructional features of synchro control transformer are similar to that of synchro transmitter, except the shape of rotor. The rotor of control transformer is umbrella-shaped so that the air gap is practically uniform. The induced emf in the control transformer is applied as input to the stator coils of control transformer. The rotor of control transformer is connected to the load whose position has to be maintained at the desired value. Depending on the difference between the rotor and the stator emf, an emf is induced on the rotor winding. This emf can be used to drive a motor so that the position of the load is corrected.

SYNCHRO AS ERROR DETECTOR

The synchro error detector is formed by the interconnection of a synchro transmitter and a synchro control transformer. In this arrangement, the stator leads of the transmitter are directly connected to the stator leads of the control transformer. The synchro transmitter has dumb-bell shaped rotor and its winding is connected to single-phase ac supply. The synchro control transformer has an umbrella-shaped rotor core. Error signal is available across the rotor winding of the control transformer.



$e(t) = E \sin \omega t$ = instantaneous value of input voltage

$e_o(t)$ = output voltage from control transformer rotor (error voltage)

Let the rotor of the transmitter rotate through an angle θ from its electrical zero. The rotor of the control transformer will rotate in the same direction through an angle α . The net angular separation of the two rotors is equal to $(90 - \theta + \alpha)$ and the voltage from the control transformer rotor is proportional to the cosine of this angle.

Output voltage can be written as, $e_o(t) = KE \cos(90 - \theta + \alpha) \sin \omega t$ where K is a constant.

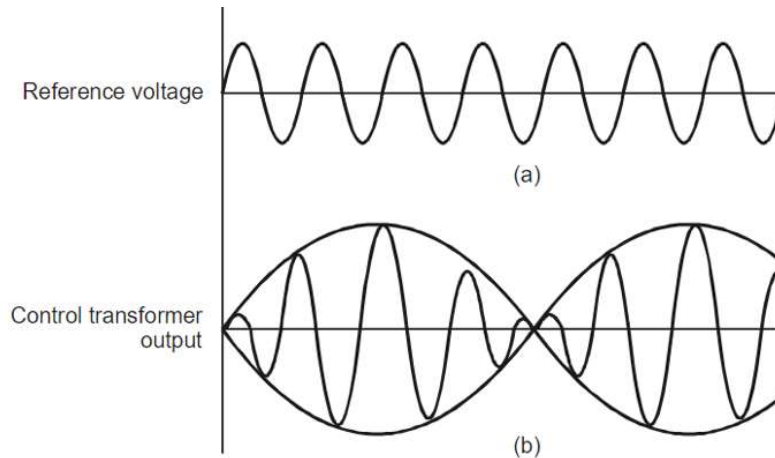
$$e_o(t) = KE \sin(\theta - \alpha) \sin \omega t$$

Output voltage $e_o(t)$ is zero when $\theta - \alpha = 0^\circ$.

Let $\varphi(t) = \theta - \alpha$

$$e_o(t) = KE \sin \varphi(t) \sin \omega t$$

For small value of $\varphi(t)$, $e_o(t) = KE \varphi(t) \sin \omega t$



Stepper motor rotates through a fixed angular step in response to each input current pulse from its controller. Stepping motors are ideally suited for situations where either precise speed control or both are required in automation systems.

The angle through the motor shaft rotates for each command pulse is called the step angle, greater the number of steps per revolution and higher the resolution and positioning obtained. The steps angles can be as small as 0.72° or as large as 90° .

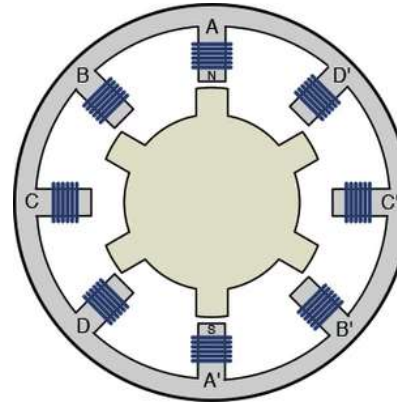
The value of step angle β can be expressed either in terms of the rotor and stator poles respectively or in terms of the number of stator phases (m) and the number of rotor poles (N_r).

$$\beta = \frac{N_s - N_r}{N_s \times N_r} \times 360^\circ = \frac{360^\circ}{mN_r}$$

TYPES OF STEPPER MOTORS

Variable reluctance stepper motor

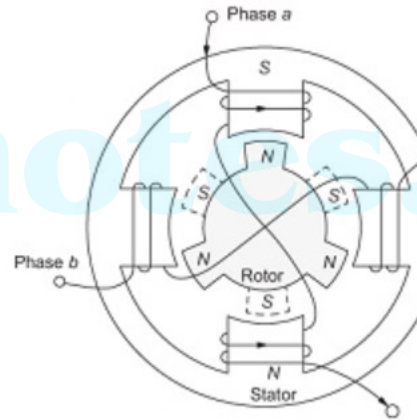
It has wound stator poles but the rotor poles are made of magnetic material as shown.



It can be of the single stack type or multi-stack type with different step angles. Direction of motor rotation is independent of the current. It is called variable reluctance motor because the magnetic circuit formed by the rotor and stator teeth varies with the position of the rotor.

Hybrid stepper motor

It has wound stator poles and permanently-magnetized hybrid motor it has a combination of advantages of both variable reluctance rotor, which mainly consists of high torque. Due to its hybrid nature, its costs and construction



VARIABLE RELUCTANCE STEPPER MOTOR

A variable-reluctance motor is constructed from ferromagnetic material with same figure. The stator is made from a stack of steel laminations and has six equally-spaced teeth (or teeth) each wound with an exciting coil. The rotor which may be solid or have projecting teeth of the same width as the stator teeth. As seen, there are three independent windings or phases A, B and C and each one can be energized by a direct current pulse from a microprocessor.

The six stator coils are connected in 2-coil groups to form three separate circuits. Each phase has its own independent switch. Diametrically opposite pairs of stator coils are energized such that when one tooth becomes a N-pole, the other one becomes a S-pole.

When there is no current in the stator coils, the rotor is completely free to rotate. The energizing of more stator coils causes the rotor to step forward (or backward) to a position where the reluctance is minimized.

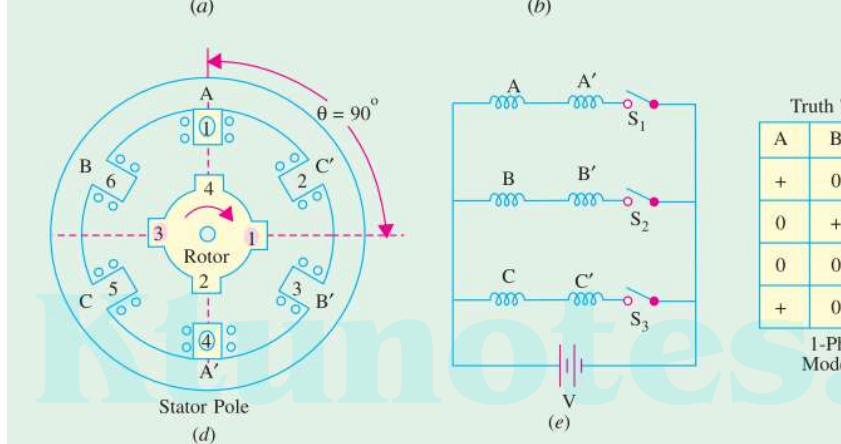
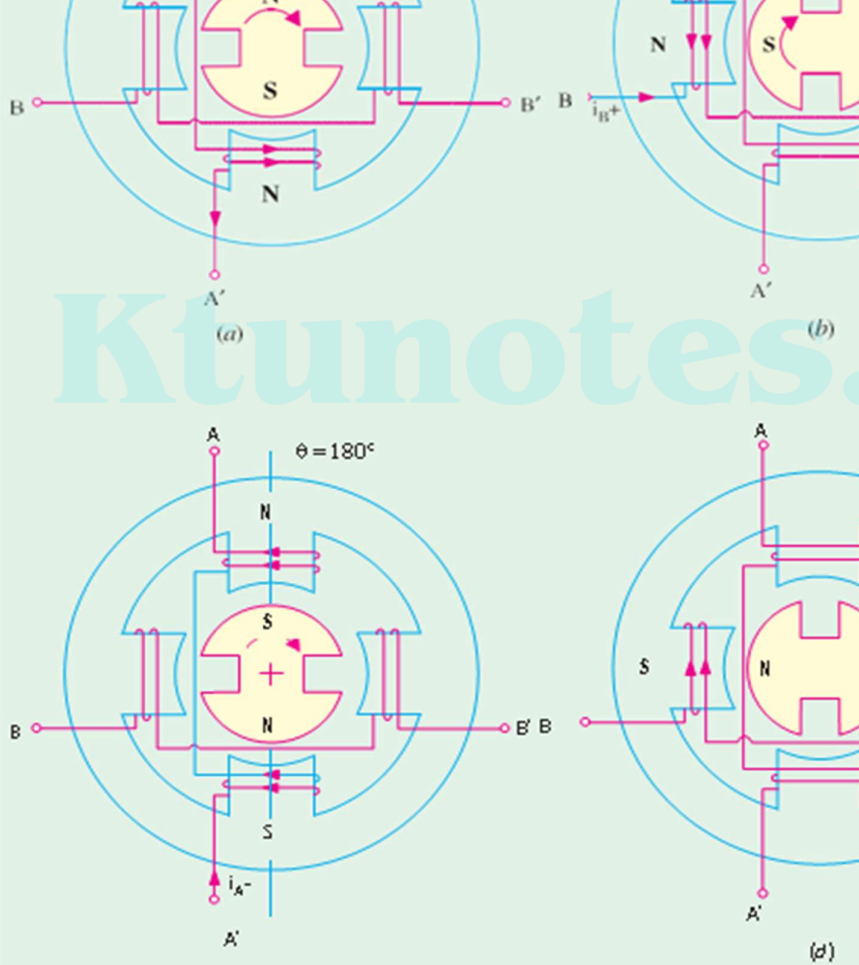


Fig. (a) shows the position of the rotor when switch S_1 has been closed for a magnetic field with its axis along the stator poles of phase A is created. The rotor turns into a position of minimum reluctance with diametrically opposite rotor teeth aligned with stator teeth 1 and 4 respectively. Closing S_2 and opening S_1 energizes phase B causing the rotor to turn through an angle of 30° in the clockwise (CW) direction. Similarly, when S_3 is closed after S_2 is opened which causes rotor teeth 1 and 3 to line up with stator teeth 2 and 5 respectively as shown in Fig. (c). The rotor rotates through an additional angle of 30° in the clockwise direction. When S_3 is opened and S_1 is closed again, the rotor teeth 2 and 4 will align with stator teeth 3 and 6 respectively thereby making the rotor turn through a further angle of 30° as shown in Fig. (d). The total angle turned is 90° .

As each switch is closed and the preceding one opened, the rotor each time rotates through an angle of 30° . By repetitively closing the switches in the sequence 1-2-3-1 and thus energizing the stator in the sequence ABCA etc., the rotor will rotate clockwise in 30° steps. If the switches are closed in the sequence 3-2-1-3 which makes phase sequence CBAC (or ACB), the rotor will rotate anticlockwise. This mode of operation is known as 1-phase-ON mode or full-step operation and is the simplest method of making the motor step.



When a particular stator phase is energized, the rotor magnetic poles move into alignment with the excited stator poles. The stator windings A and B can be excited with either positive or negative current (i.e., from i_{A+} to positive current i_{A+} in the phase A and A' to negative current i_{A-}).

Fig. (a) shows the condition when phase A is excited with positive current i_{A+} . If the current is now switched to phase B as in Fig. (b), the rotor rotates by a full step of 90° in the clockwise direction. Next, when phase A is excited with negative current i_{A-} , the rotor turns through another 90° in the clockwise direction.

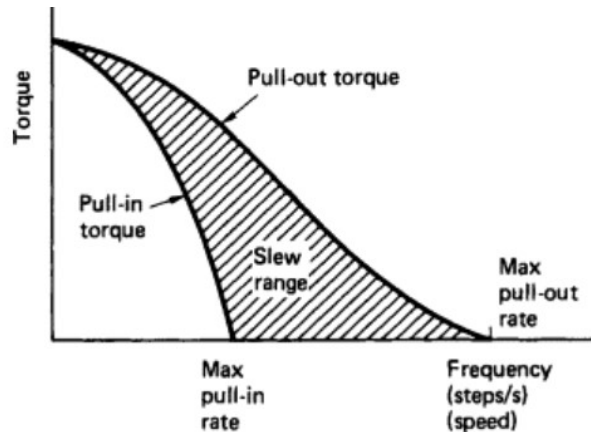
Dynamic torque: The torque developed by the motor at very slow stepping speed.

Holding torque: The maximum torque which can be applied to an energized stepper motor without causing spindle rotation.

Pull-out rate: The maximum switching rate at which a motor will remain in synchronism when the switching rate is gradually increased.

Pull-in rate: The maximum switching rate at which a loaded motor can start with a given load.

Slew range: The range of switching rates between pull-in and pull-out in which the motor will remain in synchronism but cannot start or reverse.



During the application of each sequential pulse the rotor of a stepper motor accelerates to the new step position. However, on reaching the new position there will be some oscillation unless sufficient retarding torque is provided to prevent this happening. This can cause rotor resonance at certain pulse frequencies, resulting in loss of torque and pull-out conditions. As variable-reluctance motors have very little inherent damping, they are more susceptible to resonances than either the permanent magnet or the hybrid types. If electronic dampers are available which can be used to minimize the adverse effects of resonance. If at all possible, however, the motor should be selected such that its resonant frequency is well above the application under consideration.

system

It should be less sensitive to system parameter change

It should be able to reduce the effect of undesirable disturbance

It should have as little steady-state error as possible.

NEED OF COMPENSATION: A control system is designed to perform a certain set of specifications. Sometimes, a closed loop system may not be satisfactory with respect to its characteristics, speed of response and steady-state accuracy. The system may become unstable. It may have either extremely fast or very sluggish response. The error between the actual and desired values may be excessive and not acceptable. Therefore, we need to modify the system or system parameters to provide the desired performance with respect to its characteristics. This has certain practical limitations, such as size, range, cost of components.

In such cases the performance of the control system is improved by adding a compensator. The method of improving the performance in this way is called compensation. A compensator component changes the transfer function of the overall system and gives the desired performance. In feedback systems the compensation added is simple and less expensive and results in improvement in the performance.

A controller or a compensator is a device which when introduced in forward path of a control system, controls the steady-state and transient response according to the requirements.

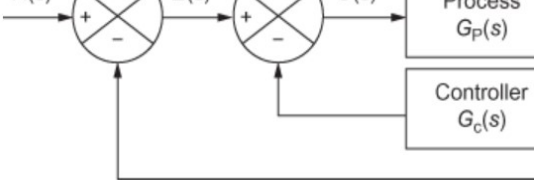
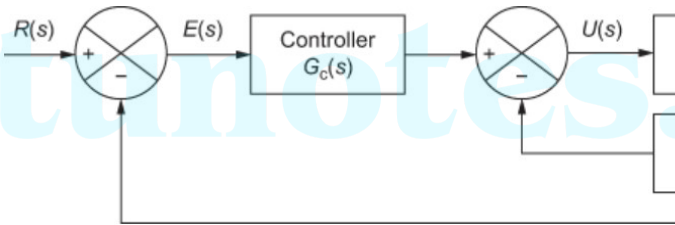
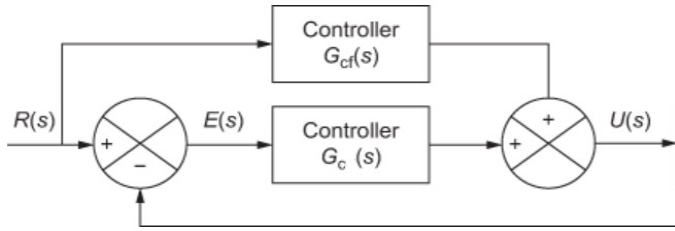
A compensator

- i) Make an unstable system stable
- ii) Minimize overshoot
- iii) Increase steady-state accuracy
- iv) Improves transient response

TYPES OF COMPENSATION

A compensator is inserted into the system at a suitable location depending on the type of compensation. Compensation is usually classified as follows:

1. Cascade (or series) compensation
2. Feedback (or parallel) compensation

Feedback (or parallel) compensation	 The addition of compensator in the feedback path increases the system that makes it accurate and more stable
Series-parallel compensation	 It is a combination of both series and parallel compensation.
Feed-forward compensation	 The feed forward controller is placed in parallel with the closed-loop system. It has a controller in the forward path.

LEAD COMPENSATOR

The lead compensator is an electrical network which produces a sinusoidal output when a sinusoidal input is applied.

$$G(s) = \frac{R_2(sR_1C + 1)}{R_1 + R_2(sR_1C + 1)}$$

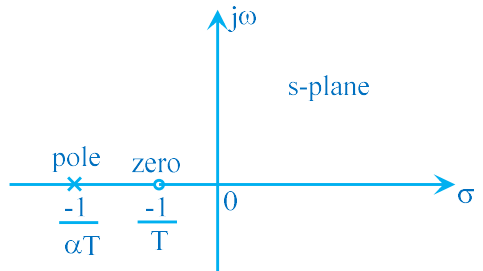
$$G(s) = \frac{R_2(sR_1C + 1)}{R_1 + R_2(sR_1C + 1)} = \frac{R_2}{R_1 + R_2} \frac{1 + sR_1C}{1 + \frac{R_2}{R_1 + R_2}(1 + sR_1C)} = \frac{\alpha(1 + sT)}{1 + \alpha sT}$$

Let $T = R_1C$ and $\alpha = \frac{R_2}{R_1 + R_2} < 1$

Dividing both numerator & denominator by αT ,

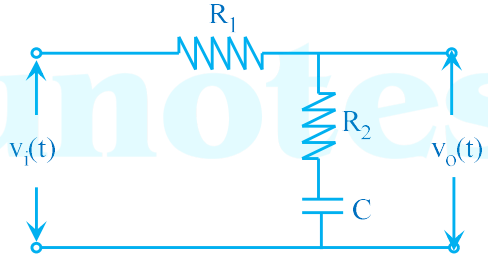
$$G(s) = \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}; \alpha < 1$$

Lead network has a zero at $s = \frac{1}{T}$ and a pole at $s = \frac{1}{\alpha T}$. Since $0 < \alpha < 1$, the zero is to the right of the pole in the complex s-plane. Since zero is always nearer to the origin than the pole, the effect of zero is dominant.



LAG COMPENSATOR

A lag compensator is a circuit that is designed to generate a steady-state sinusoidal phase lag to the applied input sinusoidal signal.



$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)} = \frac{R_2 + \frac{1}{Cs}}{R_1 + R_2 + \frac{1}{Cs}} = \frac{sR_2C + 1}{s(R_1 + R_2)C + 1}$$

$$\text{Let } T = R_2C \text{ and } \beta = \frac{R_1 + R_2}{R_2} > 1$$

$$\beta T = \left(\frac{R_1 + R_2}{R_2} \right) R_2C = (R_1 + R_2)C$$

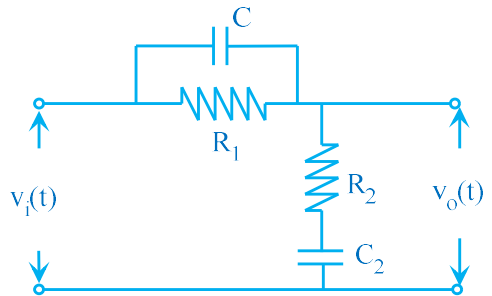
$$G(s) = \frac{sR_2C + 1}{s(R_1 + R_2)C + 1} = \frac{sT + 1}{s\beta T + 1} = \frac{1}{\beta} \times \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}}$$

Limitations of lag compensator	Bandwidth is reduced and transient response becomes slower
	More sensitive to parameters variation

LEAD-LAG COMPENSATOR

Lead compensators are employed to improve transient response. To meet the steady-state error, lag compensators are used. Only single compensator, lag or lead, can not meet both requirements simultaneously. Lag-lead compensators are used when both transient as well as steady-state error are to be improved.

Lag-Lead compensator is an electrical network which produces phase lag at low frequency (low frequency) region and phase lead at other frequency (high frequency) region. It is a combination of the lag and the lead compensators.



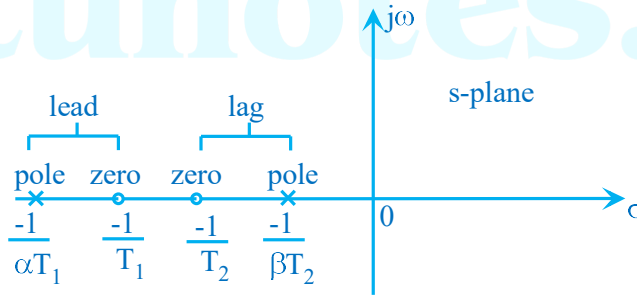
$$Z_1(s) = R_1 // \frac{1}{sC_1} = \frac{R_1 \times \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{R_1}{sR_1C_1 + 1}$$

$$Z_2(s) = R_2 + \frac{1}{sC_2} = \frac{sR_2C_2 + 1}{sC_2}$$

$$G(s) = \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)}$$

where $\beta > 1$, $R_1 C_1 = T_1$, $R_2 C_2 = T_2$, $R_1 C_1 + R_2 C_2 + R_1 C_2 = \frac{T_1}{\beta} + \beta T_2$

Ktunotes



Effects of lead-lag compensator	The lead characteristics provide improvement in and the response becomes faster.
	Due to the use of the lag network, the steady-state system gets improved.

TYPES OF CONTROLLERS

Generally, system performance can be improved by using any of the following

1. Proportional control
2. Proportional Derivative (PD) control
3. Proportional plus Integral (PI) control
4. Proportional plus Integral plus Derivative (PID) control

$$\frac{E_a(s)}{E(s)} = K_p$$

The proportional controller amplifies the error signal by an amount K_p .

Introduction of P controller increases the forward path gain by K_p . If the forward path gain is increased, the peak overshoot increases while the steady-state error is reduced. In actual systems, a trade-off between peak overshoot and steady-state errors is desired to be small. Hence, a compromise is made for which the peak overshoot and the steady-state error are within specified limits.

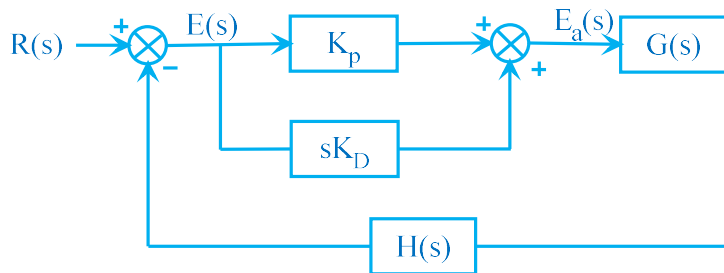
PROPORTIONAL plus DERIVATIVE (PD) CONTROLLER

The proportional plus derivative controller, the actuating signal $e_a(t)$ varies in proportion to the error signal as well as with the derivative of the error signal.

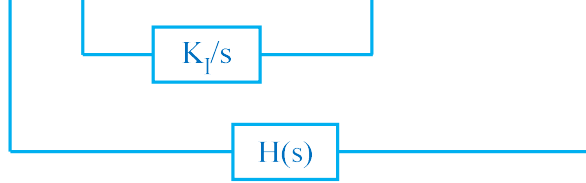
$$e_a(t) = K_p e(t) + K_D \frac{de(t)}{dt}$$

Transfer function of PD controller is

$$\frac{E_a(s)}{E(s)} = K_p + sK_D$$



The proportional derivative controller is used to improve the stability of the system by reducing the overshoot and settling time, without affecting the steady state error.



The proportional integral controller is used to decrease the steady state error and improve the stability of the control system.

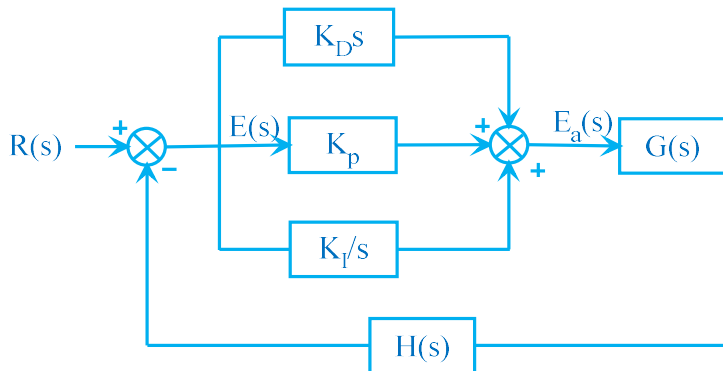
PROPORTIONAL-INTEGRAL-DERIVATIVE (PID) CONTROLLER

The proportional plus integral controller, the actuating signal $e_a(t)$ varies proportionally with the error signal $e(t)$ and also with the integral of the error signal.

$$e_a(t) = K_p e(t) + K_I \int_0^t e(t) dt + K_D \frac{de(t)}{dt}$$

Transfer function of PID controller is

$$\frac{E_a(s)}{E(s)} = K_p + \frac{K_I}{s} + K_D s$$



The proportional integral derivative controller is used to improve the stability of the control system and to decrease steady state error.